Project Report

Survival and Longitudinal Data Analysis

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Abstract

Forecasting the electricity consumption of building complexes is a challenging task due to the high frequency of the data, the presence of missing values, and the coexistence of multiple seasonalities. In this work, we conduct a comparative study of statistical and machine learning methods for modeling electricity demand in a university campus. We first address the problem of irregular sampling by applying preprocessing techniques such as gap filling, smoothing, and the construction of exogenous variables. We then evaluate several forecasting models : classical approaches such as SARIMA, MSTL or Dynamic Harmonic Regression, hybrid statistical–machine learning methods as Prophet, and neural networks like feed-forward, CNN, LSTM. Our results show that missing values and time gaps have a huge impact on the model's forecast the performance, with raw irregular data leading to significantly degraded accuracy. Among the tested models, neural networks achieve the best performance, with CNNs and LSTMs reaching RMSE values around 19–20, while statistical methods remain limited in flexibility and robustness. Prophet demonstrated competitive accuracy thanks to its ability to model calendar effects explicitly. The study highlights both the negative impact of time gaps in time series and the superiority of neural architectures in capturing complex nonlinear dynamics, while pointing to future directions such as hybrid models and methods explicitly designed for irregular sampling.

Folder containing all the code : Here

Liste des sigles et abréviations

Here is the list of abreviation used in this work :

— **SARIMA** : Seasonal Auto Regressive Integrated Moving Average
— **LOESS** : locally weighted regression
— **MSTL** : Multiple Seasonal-Trend Decomposition using LOESS
— **RMSE** : Root Mean Square Eroor
— **ETS** : Exponetial Smoothing
— **DHR** : Dynamic Harmonic Regression
— **CNNs** : Convolutional Neural Networks
— **LSTM** : Long Short Term Memory

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1 Introduction

1.1 Summary of the dataset

The METABRIC (Molecular Taxonomy of Breast Cancer International Consortium) dataset contains detailed information on patients diagnosed with primary breast cancer. The information can be categorized in 3 type : clinical, pathological, and molecular. The dataset includes variables related to patient demographics, tumor biology, treatments, molecular subtyping, and survival outcomes.

Before diving into each variable, just by looking at the dataset, we can already remove some of them because of redundancies. cancer type, Study ID, Sample Type, Number of Samples Per Patient take only one variable or are exactly the same as an other variable, so they were removed. For Patient's sex ; in this dataset, all individuals are female. Below is a structured description of each variable included in our analysis, along with relevant contextual details.

Clinical and Demographic Variables

— *Patient ID* [chr] Unique identifier assigned to each patient.

— *Age at Diagnosis* [num] Age (in years) when the patient was diagnosed with breast cancer.

— *Sex* [chr] Patient's sex ; in this dataset, all individuals are female.

Tumor Characteristics and Pathology

— *Cancer Type Detailed* [chr] Specific cancer subtype as defined in the dataset.

— *Cellularity* [chr] The percentage of tumor volume occupied by invasive tumor cells, categorized as low, moderate, or high.

— *Neoplasm Histologic Grade* [int] A histologic description of how closely the cancer cells resemble their normal cell of origin. In the breast, the overall grade score is calculated by looking at the mitotic rate, the nuclear grade or atypia, and the degree of gland formation. The final grade will be either grade 1 (low), 2 or 3 (high).

— *Tumor Other Histologic Subtype* [chr] Histologic pattern of the tumor (few missing). While histology influences tumor biology, studies show that subtype does not significantly affect 5-year overall survival in early-stage disease (Mubarak et al., 2024) (1)

— *Lymph nodes examined positive* [int] Lymph node-positive means at least one axillary lymph node contains cancer. Age less than 60, recurrence, mastectomy, larger tumor size, and shorter tumor-nipple distance were all associated with 3+ positive

lymph nodes (Elleson et al., 2023) (2).

— *Tumor Size* [int] Primary tumor diameter (in millimeters).

— *Tumor Stage* [int] Clinical stage of the tumor, ranging from 1 to 4.

— *Primary Tumor Laterality* [chr] Indicates whether the tumor was located in the right or left breast.

Molecular and Genomic Variables

— *Pam50 + Claudin-low Subtype* [chr] The PAM50 classifier assigns tumors to Luminal A, Luminal B, HER2-enriched, Basal-like, or Normal-like based on a 50-gene signature. Claudin-low is a distinct phenotype rather than a classical intrinsic subtype, but is treated as such in this dataset. (Fougner et al., 2020) (3)

— *HER2 Status* [chr] Indicates whether tumors overexpress the HER2 protein (positive or negative). HER2-positive cancers grow more aggressively but respond well to HER2-targeted therapies (4).

— *HER2 Status Measured by SNP6* [chr] can be either gain, loss or neutral.

— *ER Status* [chr] If tests show you have estrogen receptor-positive breast cancer, it means cancerous cells in your breast have high estrogen levels helping them to grow. Either positive or negative.

— *ER Status Measured by IHC* [chr] Estrogen receptor status determined using immunohistochemistry (IHC). Either positive or negative.

— *PR Status* [chr] Indicates presence of progesterone receptors in tumor cells. Either positive or negative.

— *3-Gene Classifier Subtype* [chr] A robust molecular classification method based on the expression of three key genes reflecting ER status, HER2 activity, and cellular proliferation (Haibe-Kains et al., 2012) (5)

— *Mutation Count* [int] Total number of observed mutations per tumor

— *TMB (nonsynonymous)* [int] Tumor mutational burden, defined as the number of nonsynonymous mutations per coding region.

Integrated and High-Level Molecular Features

— *Integrated Cluster* [int] A ten-class taxonomy integrating copy-number variants and gene expression, each cluster associated with distinct biological drivers and prognosis. (Young et al., 2024) (6)

— *Oncotree Code* [chr] Standardized oncology diagnostic classification.

Treatment Variables

— *Type of Breast Surgery* [chr] Indicates whether the patient underwent mastectomy or breast-conserving surgery.

— *Chemotherapy* [chr] Binary variable recording whether the patient received chemotherapy (YES/NO).

— *Hormone Therapy* [chr] Indicates whether the patient received endocrine therapy (YES/NO).

— *Radiotherapy* [chr] Indicates whether adjuvant radiotherapy was administered (YES/NO).

Menopausal and Cohort Information

— *Menopausal State* [chr] Estimated menopausal status of the patient.

— *Cohort* [int] Numerical category (1–9) representing the study batch or sub-group in METABRIC.

Prognostic Scores and Indices

— *Nottingham Prognostic Index (NPI)* [num] The Nottingham prognostic index (NPI) is used to determine prognosis following surgery for breast cancer. Its value is calculated using three pathological criteria : the size of the tumour ; the number of involved lymph nodes ; and the grade of the tumour.

Survival Outcomes

— *Overall Survival (Months)* [num] Time from diagnosis to death or last follow-up.

— *Overall Survival Status* [chr] Indicates whether the patient died (1) or was alive/censored (0).

— *Relapse Free Status (Months)* [num] Time from treatment to recurrence or relapse.

— *Relapse Free Status* [chr] Indicates whether the patient experienced recurrence (1) or remained relapse-free (0).

— *Patient's Vital Status* [chr] Categorical : alive, dead from disease, or dead from other causes.

2 Exploratory Data Analysis

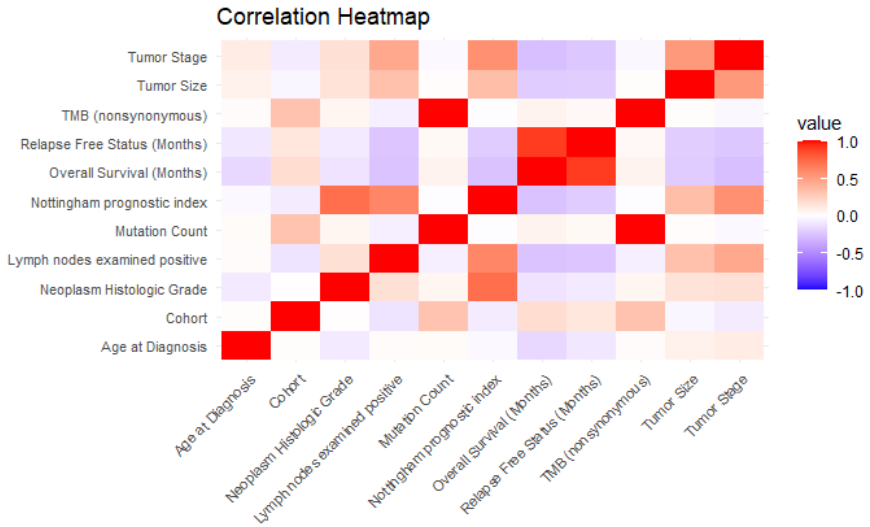
2.1 Missing values imputation

The dataset contains lots of missing data, only 3 variables has 0 missing values, while the median one has approximately 500 missing values (1/5 of the dataset). Handling these missing values is primordial.

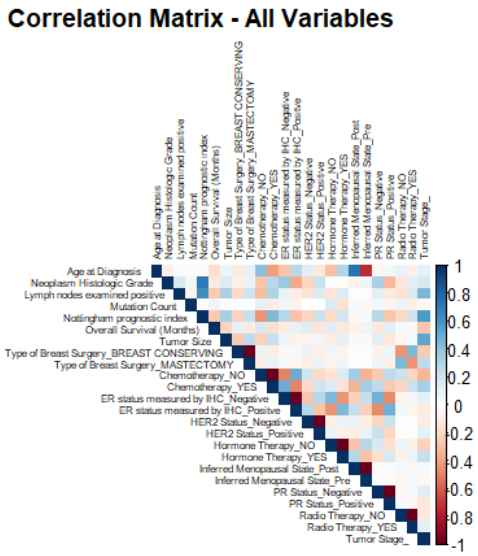
MICE Imputation, short for 'Multiple Imputation by Chained Equation' is an advanced missing data imputation technique that uses multiple iterations of Machine Learning model training to predict the missing values using known values from other features in the data as predictors.

2.2 Dependence and Correlations

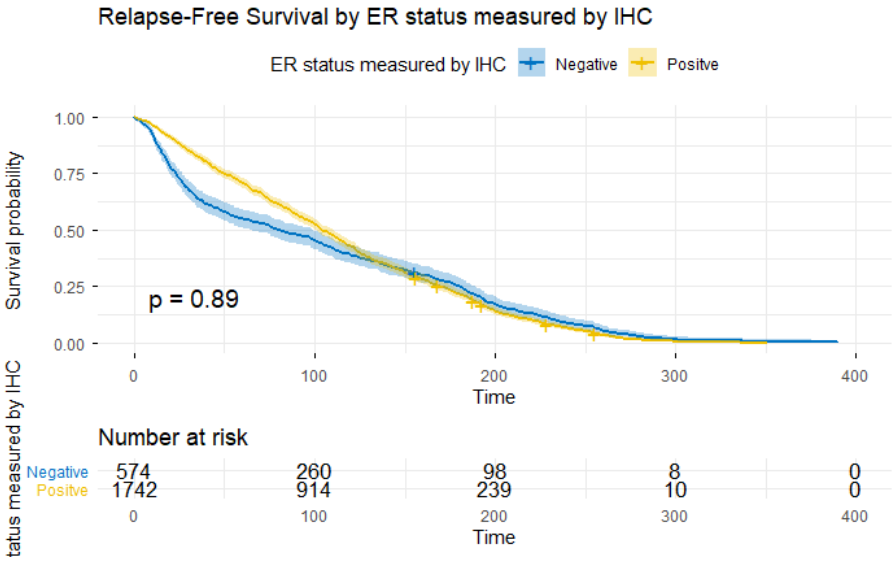
In the dataset, we observe some correlated variable. First, let's look at the continuous variables. as shown in this matrix correlation.



We see that Mutation Count and TMB (nonsynonymous) are highly correlated, which is expected since both measure mutation frequency. Nottingham Prognostic Index and Neoplasm Histologic Grade are also strongly correlated. Because of these redundancies, I removed Neoplasm Histologic Grade and Mutation Count.



After converting categorical variables to dummy variables, no other strong correlations were found. I then tested whether each categorical variable showed differences in survival across its groups. This revealed useful insights. For example, ER status (measured by IHC) showed no meaningful difference in relapse-free survival ($p = 0.89$), suggesting that ER status does not contribute predictive information in this dataset and can be removed.



Other intersting variables can be found when looking at the code. But i choose to continue to the next section in order to stay inside the text limitation.

3 Cox Models

3.1 linear Cox model

The basic Cox model overfits badly. It performs almost perfectly on the training data but does poorly on the test set, which means the model is not trustworthy. The risk groups

it produces (medium and high) are not meaningfully different from the low-risk group, so the model does not help classify patients.

This poor behavior comes from data leakage and too many parameters. Because of this, the model cannot be used to draw any real conclusions about breast cancer prognosis. A full redesign is needed, removing survival-related variables used incorrectly as predictors and reducing model complexity. The P-spline model performs better and is more reliable.

3.2 Extended Cox Model with P-splines

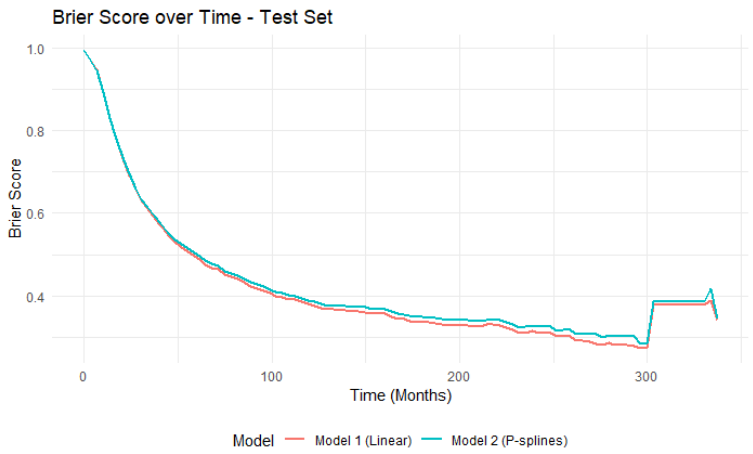
P-splines are an attractive approach for modeling nonlinear smooth effects of covariates within the additive and varying coefficient models framework. It is a non linear way of modelling that I hope will

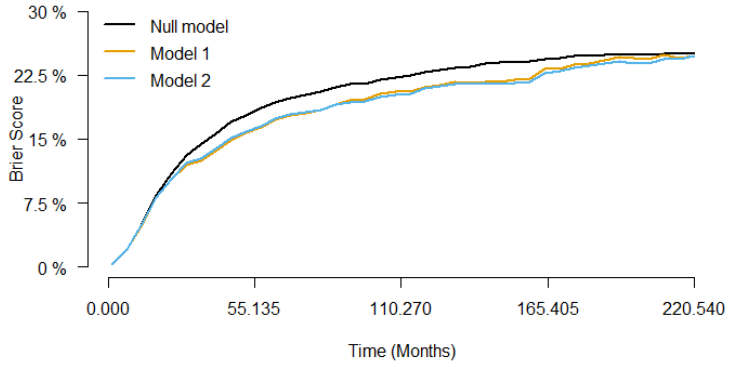
The extended Cox proportional hazards model with P-splines demonstrated moderate discriminative ability on the training dataset (C-index = 0.701, SE = 0.01), though this decreased substantially on the test data (C-index = 0.611), suggesting some degree of overfitting. The model achieved statistical significance overall (likelihood ratio test = 351.2, df = 54.79, p < 2e-16), indicating that covariates collectively provide meaningful prognostic information.

The model exhibited several features warranting interpretation :

1. substantial performance degradation from training to testing suggests overfitting, likely due to the large number of parameters (54.79 degrees of freedom) relative to the number of events (764) ;
2. extremely wide confidence intervals for rare cancer subtypes and high TMB values indicate estimation instability
3. the lack of significance for established prognostic factors like chemotherapy may reflect confounding by indication, as treatment decisions are informed by risk factors already in the model.

3.3 Brier





The Brier score reveal that both models achieve comparable probabilistic accuracy on held-out data, with meaningful improvement over baseline predictions during the clinically relevant 5-10 year follow-up window. However, the convergence with the null model at longer follow-up times indicates limitations in long-term prognostic accuracy. These findings, combined with the C-index results, suggest that while the P-spline model (Model 2) is methodologically superior and more interpretable, neither model achieves exceptional predictive performance, with most predictive information concentrated in cancer subtype classifications rather than continuous clinical or molecular variables.

3.4 Predicted Survival Curves for Representative Patient Profiles

To show how the model can be used in practice, we generated predicted survival curves for three example patient profiles.

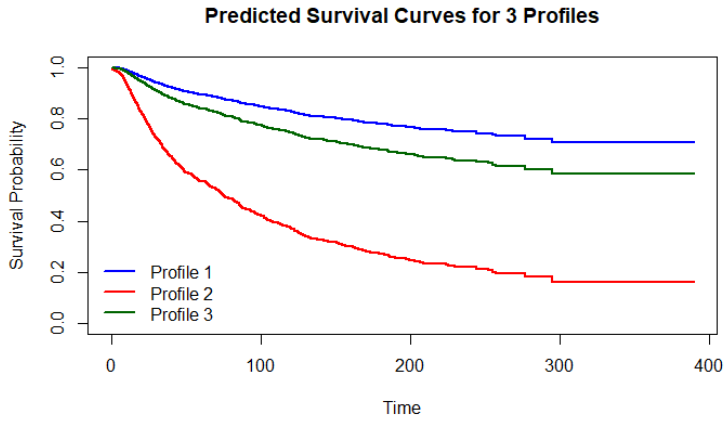
Profile 1 (low risk) : a 50-year-old with early-stage disease (15 mm tumor, stage 1, no lymph nodes involved), luminal A subtype, high cellularity, treated with hormone therapy and radiotherapy. Profile 2 (intermediate risk) : a 60-year-old with a 28 mm tumor, stage 2, 3 positive nodes, basal subtype, treated with chemotherapy only. Profile 3 (high risk) : a 75-year-old with a 45 mm tumor, stage 3, 10 positive nodes, luminal B / HER2-positive subtype, low cellularity, treated with chemotherapy and hormone therapy.

The predicted curves show clear differences between these groups. Profile 1 keeps a high survival probability, around 70% at 12 years. Profile 3 falls to about 60% at 12 years. Profile 2 has the worst outlook, dropping to roughly 15–20% at 12 years. Although Profile 3 appears clinically "highest risk," its better survival compared to Profile 2 likely reflects tumor biology : the basal subtype in Profile 2 is especially aggressive.

For conditional survival, take a patient like Profile 2 who has already survived 7 years after diagnosis. Their survival probability at that time is about 30%. The chance of surviving 5 more years (to year 12) is

$$P(T > 144|T > 84) = \frac{0.15}{0.30} = 0.50$$

. This is much higher than the unconditional 12-year probability of 15%. The reason is that survival is not memoryless : reaching 7 years without an event indicates better long-term prospects. In practice, this shows that long-term survivors often have a much improved outlook, which matters for follow-up planning and patient reassurance.



4 Conclusion

4.1 Conclusion

This project developed Cox proportional hazards models for breast cancer survival prediction using clinical and molecular features. Model 1 suffered from critical specification errors (inadvertent inclusion of outcome variables), while Model 2 achieved modest performance (test C-index = 0.61) with significant overfitting. Key prognostic factors included cancer subtypes, molecular classifications, and hormone therapy. Brier score analysis showed models provided value for 5-10 year predictions but converged with null predictions at longer follow-up.

A future idea would have been to compare the Cox model with a tree-based model using random forest, which I wasn't able to do due to computational issue.

Another interesting idea would be to look more closely at the patient's vital status. There are three categories : alive, dead from breast cancer, and dead from other causes. We could treat "dead from other causes" the same as "alive," because that death is unrelated to the outcome we are studying. Statistically, it is as if the patient were still alive for the purpose of analyzing breast-cancer–specific survival.

5 References

1. Mubarak, F., Kowkabany, G., Popp, R., Bansal, S., Ahmed, S. H., Sharan, S., Sukniam, K. B., Raikot, S. R., Jimenez, P. B., Popp, K., Manaise, H. K., & Gabriel, E. (2024). Early Stage Breast Cancer : Does Histologic Subtype (Ductal vs. Lobular) Impact 5 Year Overall Survival ? Cancers, 16(8), 1509. Link to the article

2. Elleson, K. M., Englander, K., Gallagher, J., Chintapally, N., Sun, W., Whiting, J., Mallory, M., Kiluk, J., Hoover, S., Khakpour, N., Czerniecki, B., Laronga, C., & Lee, M. C. (2023). Factors Predictive of Positive Lymph Nodes for Breast Cancer. Current Oncology, 30(12), 10351-10362. Link to the article

3. Fougner, C., Bergholtz, H., Norum, J. H., & Sørlie, T. (2020). Re-definition of claudin-low as a breast cancer phenotype. Nature Communications, 11(1), 1787. Link to the article

4. cancer.org

5. Haibe-Kains, B., Desmedt, C., Loi, S., Culhane, A. C., Bontempi, G., Quackenbush, J., & Sotiriou, C. (2012). A Three-Gene Model to Robustly Identify Breast Cancer Molecular Subtypes. JNCI Journal Of The National Cancer Institute, 104(4), 311-325. Link to the article

6. Young, C. C., Eason, K., Garcia, R. M., Moulange, R., Mukherjee, S., Chin, S., Caldas, C., & Rueda, O. M. (2024). Development and validation of a reliable DNA copy-number-based machine learning algorithm (CopyClust) for breast cancer integrative cluster classification. Scientific Reports, 14(1), 11861. Link to the article

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